Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

PROJECT TITLE : HYBRID NNP CHECKER

TEAM MEMBERS :

- **VISHVA S A [23MIS0420]**
- **DHANUSH T[23MIS0404]**
- **DHANUSH M[23MIS0526]**

PANEL NO.: 01

VENUE : PRP 315R

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Problem Statement:

Board games like Checkers are valued for their tactile interaction, cultural heritage, and social bonding, but they face major limitations in today's connected world. Traditional play requires both players to be physically co-located, while digital versions sacrifice the authentic hands-on experience of moving real pieces. Current hybrid attempts often rely on manual input or partial automation, leading to inconvenience, inconsistency, and reduced immersion.

Project Abstract :

Hybrid Checkers – Play Anywhere, Stay Connected is a smart board game that merges physical play with remote and AI modes. Hall sensors detect moves and send them to the cloud, while an XY gantry with electromagnets replicates them on the opponent's board, enabling "you move here, it moves there" gameplay. The system supports Physical↔Physical, Physical↔AI, and Physical↔Web play with real-time sync, AI levels, and a web interface, ensuring accuracy, immersion, and commercialization potential.

Proposed Solution and Innovation :

- A hybrid Checkers platform with Hall sensors under each square to detect the local player's moves.
- A microcontroller maps moves to coordinates and pushes them to the cloud.
- The opponent's board uses a CNC-style XY gantry with electromagnets under the board to automatically move opponent pieces (pick/drag/step).
- Supports three modes: Physical↔Physical, Physical↔AI, and Physical↔Web.
- Built-in safety (limit switches, soft limits), calibration routine, and fallback to manual mode.
- Cloud sync ensures both boards (or board+web) stay consistent with millisecond-level updates.

Budget & Fund Utilization

1. Hardware Components

- Hall Sensors (64 pcs) → ₹2,500 – ₹3,000
- Microcontroller (ESP32/Arduino Mega) → ₹1,200 – ₹2,000
- XY Gantry (CNC rails, stepper motors, belts, drivers) → ₹8,000 – ₹10,000
- Electromagnets (2–3 units) → ₹2,000 – ₹3,000
- Power Supply, Limit Switches, Wiring, PCB, Connectors → ₹3,000 – ₹4,000
- Checkers Board & Pieces → ₹1,000 – ₹1,500

2. Software & Cloud

- Cloud Hosting (Firebase/MQTT, 3 months) → ₹2,000 – ₹3,000
- Web App Development → ₹5,000 (outsourced) / NIL (in-house)
- AI Engine Integration → Free (open-source)

3. Prototyping & Fabrication

- 3D Printing / CNC machining → ₹3,000 – ₹5,000
- Assembly Tools & Miscellaneous → ₹2,000 – ₹3,000

4. Testing & Demonstration

- Spare Parts (motors, magnets, sensors) → ₹3,000 – ₹4,000
- Demo Setup (presentation kit, posters) → ₹2,000 – ₹3,000